

Human and LLM Negotiation Dynamics: Dialogue Acts, Utility, and Agreement Evaluation

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Abstract

This project investigates how Large Language Models behave as negotiation agents. The analysis uses the Negochat corpus as a reference for human-produced negotiation language and compares it with several LLM-based negotiation experiments. The project focuses on dialogue acts, negotiated issues, inferred outcomes, agreement evaluation rules, and utility-based agreement validity. The central finding is that LLM agents can generate locally plausible negotiation behavior, such as offers, counteroffers, questions, and acceptance statements. However, their ability to reach a stable and valid agreement is fragile and depends strongly on how agreement is defined by the experimental environment. In particular, explicit acceptance rules produce many timeouts, compatibility-based rules overestimate agreement, and utility-based evaluation reveals that explicit acceptance is not always strategically valid. Nonetheless, LLMs show negotiation fluency, but weak and evaluation-sensitive negotiation closure.

1 Introduction

Negotiation is a complex form of dialogue in which participants exchange offers, ask questions, and gradually move toward agreement or failure. In human communication, negotiation requires both local conversational ability and global strategic reasoning. A negotiator must not only produce plausible conversations, but also remember goals, constraints, concessions, and the conditions under which an offer should be accepted. This project studies negotiation as an agentic behavior of Large Language Models (LLMs). The goal is to understand what happens when LLMs are placed in negotiation environments and asked to behave as autonomous agents. In particular, the project asks whether LLMs can reproduce human-like negotiation moves, whether they can reach agreement, and how their apparent success depends on the evaluation rules used by the environment. Two or more LLM agents are placed in simulated negotiation settings where they must exchange offers and attempt to reach agreement.

The main research question is:

How do LLM agents behave in negotiation settings compared with human negotiators, and how sensitive are their outcomes to prompting, utility constraints, and termination rules?

The goal is not to build a perfect negotiation model. Instead, the project aims to understand what happens when LLMs are treated as negotiation agents. In particular, I investigate whether LLMs can reproduce human-like negotiation moves and how much of their success depends on how agreement is defined by the experimental environment.

2 Dataset and Human Baseline

The human baseline is based on the Negochat corpus, an annotated natural language dialogue corpus in the negotiation domain. Negochat contains 105 job negotiation dialogues between an employer and a

candidate. The dialogues are annotated with negotiation dialogue acts such as Offer, Accept, Reject, Query, Greet, and Quit. In this project, the raw JSON files are then converted into a structured dataframe with one row per annotated dialogue act and the processed dataframe contains 4993 rows. Each row contains information such as the dialogue ID, turn ID, speaker role, text, dialogue act, issue, and value.

2.1 Wizard-of-Oz Setup

The Negochat corpus was collected using a Wizard-of-Oz methodology, meaning that human-produced negotiation utterances were mediated and mapped into structured semantic annotations by a hidden operator. Consequently, the dataset represents a human-agent interaction baseline rather than an organic human-human dialogue. This setup provides high-quality natural language utterances paired with reliable dialogue-act representations, making it an ideal structured benchmark for evaluating autonomous LLM counterparts.

2.2 Dialogue Acts and Issues

Each dialogue consists of phrases annotated with dialogue acts (*Offer*, *Accept*, *Reject*, *Query*, *Greet*, *Quit*) that represent the communicative function of the utterance. It is crucial to distinguish these conversational actions from the underlying negotiation issues (the contract dimensions being discussed). For example, in the utterance “I can offer 85,000 dollars”, the dialogue act is *Offer*, while the issue is *Salary* and its value is *85,000*.

2.3 Dialogue Act and Issues Distribution

The distribution of dialogue acts in the human corpus shows that negotiation is mainly structured around offers and counteroffers. Table 1 reports the dialogue act counts.

Dialogue act	Count
Offer	2996
Reject	790
Accept	547
Other	394
Greet	135
Query	113
Quit	18

Table 1: Dialogue act distribution in the Negochat human corpus.

This distribution confirms that the dataset is strongly negotiation-oriented: most annotations are offers, followed by rejections and acceptances.

Table 2 defines the contract issues represented in the corpus, while Figure 1 shows how frequently they appear in the annotations.

Issue	Example
Job description	Programmer, Project Manager, or Team Manager.
Salary	86.500, 75.000, 120.000.
Working hours	9 hours / 8 hours / 10 hours
Pension fund	percentage
Leased car	Boolean
Promotion possibilities	Fast / Moderate / Slow

Table 2: Negotiation issues in Negochat.

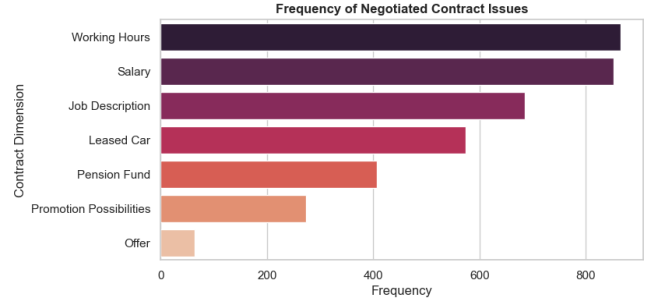


Figure 1: Distribution of negotiated issues in Negochat.

2.4 Dataset Analysis

The processed Negochat dataframe does not contain a direct final-outcome column. Therefore, human outcomes are inferred from dialogue acts using a simple heuristic. A dialogue is classified as *Agreement* if it contains at least one *Accept* act, *Failure* if it contains a *Quit* act and no *Accept* act, and *Unresolved* otherwise. Using this simple heuristic, 103 out of 105 dialogues are classified as *Agreement*, while 1 dialogue is classified as *Failure* and 1 as *Unresolved*. This heuristic is intentionally simple and is used in the rest of the analysis in order to keep a single consistent outcome definition across the human baseline and the LLM comparisons.

However, this simple criterion may overestimate final agreement, because an *Accept* act can sometimes refer to a partial issue-level acceptance rather than the final acceptance of the whole negotiation. For this reason, I also computed a refined diagnostic heuristic based on the position of the *Accept* act in the dialogue. If an *Accept* act occurs in the final turns, the dialogue is classified as *FinalAgreement*; if an *Accept* act occurs earlier but not near the end, it is classified as *PartialOrIntermediateAccept*.

This diagnostic check shows that, among the 103 dialogues classified as *Agreement* by the simple heuristic, 78 show stronger evidence of final agreement, while 25 contain an accept act that is more likely to represent a partial or intermediate acceptance. Therefore, the simple heuristic is kept as the main outcome definition for consistency, while the refined heuristic is used to qualify and interpret the human baseline for demonstration purposes.

Refined diagnostic outcome	Number of dialogues
FinalAgreement	78
PartialOrIntermediateAccept	25
Failure	1
Unresolved	1

Table 3: Diagnostic based on the position of Accept acts.

Nevertheless, the simple heuristic provides a useful approximate baseline.

3 LLM Experimental Setup

The project contains three LLM-based components. Each component answers a different question and should not be interpreted as the same type of experiment.

1. **Human-seeded LLM continuation:** This experiment is a human-seeded continuation experiment that tests whether LLM agents can continue from real human negotiation acts. In this setting, the model starts from the first substantive negotiation act of a sampled Negochat dialogue. It is used to study continuation behavior and outcome tendencies.

2. **Controlled explicit vs compatibility evaluation:** This controlled LLM experiment tests how different agreement evaluation rules affect the observed outcome. The same controlled setting was evaluated under two rules: explicit acceptance and compatibility-based agreement.
3. **Utility-based multi-issue negotiation:** This experiment requires agents to negotiate a complete contract package this means that it extends the negotiation beyond salary and working hours. Each accepted package is then evaluated through different utility measures.

3.1 Technical Setup and Reproducibility

The technical implementation details, including the Colab generation notebooks, model choices, result files, and reproducibility instructions, are documented in the project README. The Colab notebooks were used as generation notebooks for the LLM simulations. Since LLM generation is slow and hardware-dependent, the generated turns and outcomes were saved as CSV files. The final analysis is reproducible from the stored CSV files in the results directory. The LLM simulations were run using Llama-family models through Ollama on Google Colab. Smaller models were used when larger models were too slow for the available computational resources. The focus of the project is not direct model comparison, but analysis of LLM negotiation behavior under different evaluation settings.

4 1st Experiment: Human-seeded LLM Continuation

The first experiment investigates whether an LLM can continue a negotiation trajectory starting from a real human dialogue. This experiment uses the Negochat corpus as a human baseline and tests whether the model is able to preserve the direction of an ongoing negotiation when initialized from a real conversational context.

For each sampled Negochat dialogue, the first substantive negotiation act was extracted. Greetings and purely introductory turns were excluded, so that the continuation started from a meaningful negotiation move. The LLM was then prompted to continue the negotiation using a structured output format containing the following fields: message, dialogue act, issue, value, and decision. This structure was used to make the generated turns easier to parse and compare with the original human trajectories.

The experiment was run on a sample of 20 dialogues. Table 4 reports the outcomes produced by the LLM continuations.

LLM outcome	Count
Impasse	11
Agreement	9

Table 4: Outcomes in the human-seeded LLM continuation experiment.

The results show that the LLM reached agreement in 9 out of 20 cases, while 11 continuations ended in impasse. This suggests that the model was often able to produce coherent negotiation turns, but it did not consistently move the dialogue toward successful closure.

To better understand this behavior, the generated continuations were compared with the original human outcomes for the same dialogue IDs. The comparison uses the simple human outcome classification adopted in the rest of the project. Since the human dialogues are complete negotiations while the LLM outputs are continuations from selected dialogue points, the comparison should not be interpreted as a direct one-to-one performance evaluation. Instead, it is used as a qualitative diagnostic of whether the LLM preserves or changes the original negotiation trajectory.

Dialogue ID	Human outcome	LLM outcome	Human turns	LLM turns
2	Agreement	Agreement	16	13
17	Agreement	Impasse	48	21
53	Agreement	Agreement	55	10
72	Failure	Impasse	12	13
78	Agreement	Agreement	47	21
82	Agreement	Impasse	29	21

Table 5: Selected paired examples from the human-seeded LLM continuation experiment.

Table 5 shows representative paired cases. Some dialogues that originally reached human agreement were also completed as agreements by the LLM, as in dialogues 2, 53, and 78. However, other dialogues that originally reached agreement shifted to impasse in the LLM continuation, as in dialogues 17 and 82. This indicates that the model can sometimes follow the original negotiation trajectory, but it is not reliable in preserving it across cases.

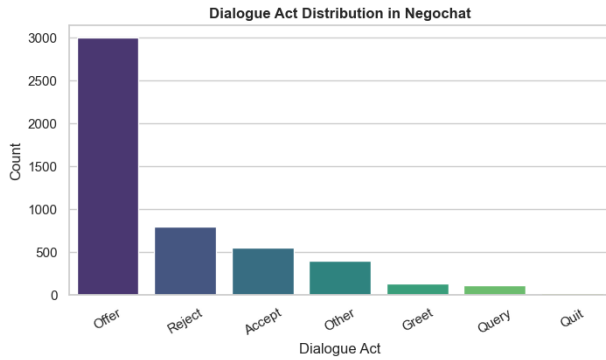


Figure 2: Dialogue-act distribution

The same pattern is visible in the dialogue-act distribution. Both the human dialogues and the LLM continuations contain a high proportion of offers. However, the LLM produces substantially fewer accept and reject acts, and a much larger proportion of queries. This suggests that the model tends to continue information-seeking behavior instead of making clear acceptance or rejection moves. As a result, the generated negotiation often remains locally plausible but fails to reach a stable final decision. Overall, the human-seeded continuation experiment shows that LLM agents can generate coherent local continuations

from real negotiation contexts. However, they are less reliable in managing the global trajectory of the negotiation, especially when successful closure requires a decisive acceptance or rejection act.

5 Evaluation Rule Sensitivity

A central methodological problem in LLM negotiation is the definition of agreement. During development, the controlled LLM experiment was evaluated using two different rules. The first rule used an explicit acceptance rule: a negotiation was classified as *Agreement* only if the model explicitly produced **DECISION: accept**. This rule is strict because it requires explicit conversational closure. The second rule was a compatibility-based evaluator. Under this rule, a negotiation is classified as *Agreement* when an offer satisfied both agents’ hard constraints. This rule is more permissive because it counts compatible offers as agreements even when the agents *do not explicitly* close the dialogue. The same type of controlled negotiation produces very different results under these two rules.

The two rules produced very different outcome distributions.

Evaluation rule	Runs	Main outcome	Interpretation
Explicit acceptance only	27	Timeout	Too strict: agents repeatedly continue bargaining and do not explicitly close.
Compatibility-based evaluator	27	Agreement	More permissive: compatible offers are counted as agreements.

Table 6: Controlled LLM outcomes under different agreement evaluation rules.

Under the explicit rule, all runs ended in timeout. This does not necessarily mean that the model never produced acceptable offers. Rather, it means that the model continued bargaining and rarely produced an explicit terminal decision.

Under the compatibility-based rule, most runs were classified as agreement. However, this rule is more permissive because it treats compatible offers as agreements even when the agents do not explicitly close the dialogue. More specifically, the explicit rule produced 27 timeouts out of 27 runs. The compatibility rule produced 25 agreements and 2 impasses out of 27 runs. This contrast suggests that LLM agents are not always able to close a negotiation autonomously. When agreement is defined only through explicit acceptance, the dialogues systematically continue until timeout. In contrast, when the environment is allowed to detect compatible offers, many negotiations can be interpreted as successful agreements. This indicates that the models may produce mutually acceptable proposals, but they often require an external evaluation mechanism to recognize closure.

5.1 Long-run Diagnostic

A long-run diagnostic was also used during development. The purpose was to test whether increasing the turn limit would solve the closure problem. The diagnostic consisted of a single longer run and was not treated as a main experimental condition. It showed that more turns can sometimes allow agreement, but longer interaction does not by itself solve the general evaluation problem. LLM agents may continue bargaining in loops or close only under particular evaluation assumptions.

6 Utility-based Multi-issue Experiment

The utility-based experiment was designed to address a limitation of simple outcome labels. In this setting, the agents negotiate a full multi-issue job contract package. A *package* is a complete job contract proposal containing values for all negotiated issues: job description, salary, working hours, pension fund, leased car, and promotion possibilities. Each proposed package is scored using private utility tables for Candidate and Employer. This experiment asks a different question from the human-seeded experiment. The question is not whether the model can continue a real dialogue context, but whether explicit acceptance corresponds to a package that is mutually beneficial according to BOTH utility scores.

Unlike the previous experiments, where the focus was mainly on whether the dialogue ended in agreement, this experiment evaluates the content of the agreement itself.

Each agent is associated with a private utility table. This table assigns a utility score to each possible value of each issue from that agent’s perspective. For instance, the candidate and the employer may assign different utilities to the same salary level or working-hours option. The utility of a full package is obtained by summing the scores of the selected values across all issues.

This allows each final proposal to be evaluated from both perspectives: candidate utility and employer utility. As a result, the experiment can distinguish between an explicit agreement and a genuinely useful agreement. A proposal may be accepted in the dialogue, but still be weak if it gives low utility to one of the agents (so this means that if you’re offering me something that I consider not good, normally I wouldn’t accept)

Overall, this experiment tests whether LLM agents can move beyond local conversational coherence and produce complete, mutually acceptable multi-issue agreements.

The Candidate generally preferred higher salary, fewer working hours, better benefits, and stronger promotion possibilities. The Employer generally preferred lower salary, more working hours, and lower-cost benefits.

For every proposed package, two scores were computed: Candidate utility and Employer utility. A valid agreement required both utilities to exceed the chosen utility threshold. This allowed the system to distinguish between linguistic acceptance and strategically valid agreement.

The utility-based experiment produced the following outcome distribution:

Utility-based outcome	Count
AcceptedLowUtility	26
Impasse	1

Table 7: Outcomes in the utility-based multi-issue LLM experiment.

In this experiment, I distinguish between two types of successful closure: *Agreement* and *AcceptedLowUtility*. An *Agreement* occurs when the agents explicitly accept a complete package that provides acceptable utility for both parties. In contrast, *AcceptedLowUtility* occurs when the dialogue contains an explicit acceptance, but the accepted package gives low utility to at least one of the agents. Therefore, both outcomes indicate that the negotiation reached some form of closure, but they have different meanings: *Agreement* represents a mutually acceptable deal, while *AcceptedLowUtility* represents a formal acceptance of a weak or unbalanced package. The dominant outcome was *AcceptedLowUtility*. This means that the model produced an explicit acceptance decision, but the accepted package did not satisfy the minimum utility thresholds for both agents. This is one of the most important findings of the project: saying “accept” is not the same as reaching a strategically valid agreement.

It shows that an LLM can perform the linguistic act of accepting an offer but can’t reproduce and recognize a strategic valid agreement. In other words, surface-level conversational closure and utility-based negotiation success are not equivalent.

7 Overall Comparison

Table 8 summarizes the LLM-based experiments.

Experiment	Brief description	Runs	Main outcome
Human-seeded LLM	LLM continuations starting from real human negotiation acts.	20	Impasse
Controlled LLM: explicit acceptance	Synthetic negotiations counted as successful only with explicit final acceptance.	27	Timeout
Controlled LLM: compatibility evaluator	Synthetic negotiations where compatible offers are counted as agreements.	27	Agreement
Utility-based LLM	Multi-issue contract negotiations evaluated with private utility tables.	27	AcceptedLowUtility

Table 8: Overall comparison of the LLM-based experiments considered in the project.

8 Discussion

The empirical findings demonstrate a sharp decoupling between **local pragmatic competence** (generating contextually appropriate turns and offers) and **global strategic closure** (steering a negotiation toward a stable, economically viable conclusion).

As summarized in Table 8, the apparent success of an LLM agent is highly sensitive to the evaluation metrics defined by the environment:

- **Explicit Acceptance:** Left to their own devices, agents fall into infinite bargaining loops, resulting in a 100% timeout rate (27/27 runs). The models mimic negotiation mechanics but lack the meta-cognitive awareness required to actively close the interaction.
- **Compatibility-Based Evaluation:** When checking only for overlapping hard constraints, the success rate spikes artificially (25/27 agreements). This metric overestimates agent cooperation, registering success even when the dialogue text remains conversational and unresolved.

- **Utility-Based Evaluation:** Introducing private utility tables exposes a critical vulnerability: linguistic alignment does not equal strategic alignment. In 26/27 runs, agents executed the conversational act of acceptance (**DECISION: accept**) but signed off on packages classified as *AcceptedLowUtility*, acting against their own prompted self-interest.

The human-seeded experiment confirms this divergence. When initialized with genuine human prompts from Negochat, LLMs failed to sustain cooperative trajectories, lapsing into impasse in 11/20 cases. The dialogue-act distribution (Figure 2) highlights the linguistic root: humans transition smoothly to definitive *Accept/Reject* acts, whereas LLMs rely heavily on information-seeking *Queries*, treating negotiation as an open-ended conversation rather than a goal-directed optimization problem. Interestingly, the experimental results remained highly uniform across the cooperative, competitive, and mixed prompting conditions. This insensitivity suggests that the behavioral limitations of the LLM agents—such as signing economically disadvantageous packages—are structural rather than stylistic, as the models failed to systematically alter their strategic convergence regardless of the assigned behavioral persona.

Limitations First, Negochat uses a Wizard-of-Oz methodology, representing a human-agent hybrid interaction rather than organic human-to-human dialogue. Second, human agreement was inferred using a simplified state-based heuristic; although the refined check (Table 3) validated 78/103 cases, the remaining 25 intermediate acceptances introduce minor noise. Lastly, computational constraints bound the simulations to smaller Llama models via Ollama, limiting the behavioral search space.

9 Conclusion

This project explored the boundary between language generation and strategic agentic behavior in LLMs. We conclude that while current LLMs excel at simulating the linguistic surface structure of negotiation, their final outcomes remain highly fragile and evaluation-sensitive. Relying on simple token alignment or raw constraint compatibility is misleading; true agentic capability must be validated by cross-referencing linguistic actions with underlying utility matrices.

AI Usage Disclaimer

Generative AI tools (including OpenAI’s ChatGPT and Gemini) were used during the development of this project to assist in project ideation, methodology workflow structuring, code debugging, and the drafting of descriptive text. All AI-generated outputs were meticulously reviewed, verified, and integrated by the author, who assumes full responsibility for the academic integrity and accuracy of the final content.